

Ex Evaluate $\iint_R e^{x^2+y^2} dy dx$

Where R is the Semicircular region bounded by the x -axis and the curve

$$y = \sqrt{1-x^2}$$

Sol

$$y = \sqrt{1-x^2}$$

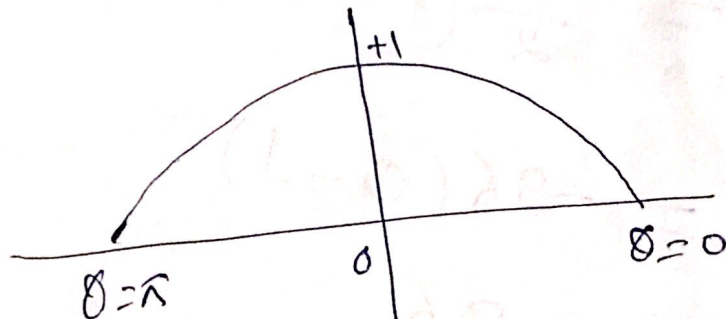
$$y^2 = 1-x^2$$

$$y^2 + x^2 = 1$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$r^2 = x^2 + y^2$$



$$\iint_R e^{x^2+y^2} dy dx = \int_0^{\pi} \int_0^1 e^{r^2} r dr d\theta$$

$$= \frac{1}{2} \int_0^{\pi} \int_0^1 e^{r^2} (2r) dr d\theta$$

$$= \frac{1}{2} \int_0^{\pi} e^r \Big|_0^1 d\theta$$

$$= \frac{1}{2} \int_0^{\pi} (e^i - e^0) d\theta$$

$$= \frac{1}{2} (e^i - 1) \theta \Big|_0^{\pi}$$

$$= \frac{1}{2} (e - 1)(\pi - 0)$$

$$= \frac{\pi}{2} (e - 1)$$

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Triple Integrals in Rectangular Coordinates ⁵⁰

$$\iiint_D f(x, y, z) dV = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k, y_k, z_k) \Delta V_k$$

Properties of Triple Integrals

$$1) \iiint_D k f dV = k \iiint_D f dV \quad (\text{any number } k)$$

$$2) \iiint_D (f \pm g) dV = \iiint_D f dV \pm \iiint_D g dV$$

$$3) \iiint_D f dV \geq 0 \quad \text{if } f \geq 0 \text{ on } D$$

$$4) \iiint_D f dV \geq \iiint_D g dV \quad \text{if } f \geq g \text{ on } D$$

$$5) \iiint_D f dV = \iiint_{D_1} f dV + \iiint_{D_2} f dV + \dots +$$

$$\iiint_{D_n} f dV$$

The Volume of a closed bounded region D in space is

$$V = \iiint_D dV$$

Ex Find the Volume of the region D enclosed by the Surfaces

$$z = x^2 + 3y^2 \text{ and } z = 8 - x^2 - y^2$$

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Sol

$$x^2 + 3y^2 = 8 - x^2 - y^2$$

$$2x^2 + 4y^2 - 8 = 0 \rightarrow (1)$$

$$4y^2 = 8 - 2x^2$$

$$y = \pm \sqrt{\frac{8 - 2x^2}{4}} = \pm \sqrt{\frac{4 - x^2}{2}}$$

For the limit of x Put $y = 0$ in eqn (1)

$$2x^2 = 8$$

$$x^2 = 4$$

$$x = \pm 2$$

$$V = \iiint dz dy dx$$

$$= \int_{-2}^{+2} \int_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} \int_{x^2+3y^2}^{8-x^2-y^2} dz dy dx$$

$$= \int_{-2}^{+2} \int_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} z \Big|_{x^2+3y^2}^{8-x^2-y^2} dy dx$$

$$= \int_{-2}^{+2} \int_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} (8-x^2-y^2-x^2-3y^2) dy dx$$

$$= \int_{-2}^{+2} \int_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} (8-2x^2-4y^2) dy dx$$

$$= \int_{-2}^{+2} \left(8y - 2x^2y - \frac{4y^3}{3} \right) \Big|_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} dx$$

$$= \int_{-2}^{+2} \left((8-2x^2)y - \frac{4}{3}y^3 \right) \Big|_{-\sqrt{\frac{4-x^2}{2}}}^{+\sqrt{\frac{4-x^2}{2}}} dx$$

$$= \int_{-2}^{+2} \left((8-2x^2) \sqrt{\frac{4-x^2}{2}} + (8-2x^2) \sqrt{\frac{4-x^2}{2}} + \left(-\frac{4}{3} \left(\frac{4-x^2}{2} \right)^{3/2} - \frac{4}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right) \right) dx$$

$$= \int_{-2}^{+2} \left(2(8-2x^2) \sqrt{\frac{4-x^2}{2}} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right) dx$$

$$= \int_{-2}^{+2} \left(4(4-x^2) \sqrt{\frac{4-x^2}{2}} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right) dx$$

$$= \int_{-2}^{+2} \left(\frac{4 \times 2 \times (4-x^2)}{2} \times \sqrt{\frac{4-x^2}{2}} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right) dx$$

$$= \int_{-2}^{+2} \left(8 \left(\frac{4-x^2}{2} \right)^{3/2} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right) dx$$

$$= \frac{16}{3} \int_{-2}^{+2} \frac{(4-x^2)^{3/2}}{\sqrt{2} \sqrt{2} \sqrt{2}} dx$$

$$= \frac{16 \sqrt{2} \times \sqrt{2} \times \sqrt{2} \times \sqrt{2} \times 4}{3 \sqrt{2} \times \sqrt{2} \times \sqrt{2}} \int_{-2}^{+2} (4-x^2)^{3/2} dx$$

$$= \frac{4\sqrt{2}}{3} \int_{-2}^{+2} (4-x^2)^{3/2} dx$$

$$= 8\pi\sqrt{2}$$

$$\frac{4\sqrt{2}}{3} \times 6\pi = 8\pi\sqrt{2}$$

$$\int_{-2}^{+2} (4-x^2)^{3/2} dx$$

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Put $x = 2 \sin \theta$

$dx = 2 \cos \theta$

$x \rightarrow -2, \theta \rightarrow -\pi/2$

$x \rightarrow 2, \theta \rightarrow \pi/2$

$$= \int_{-\pi/2}^{\pi/2} (4 - 4 \sin^2 \theta)^{3/2} 2 \cos \theta d\theta$$

$3/2 \times 2 \times 3/2 \times 3$
 $4 = 2^2 = 2^3$

$$= 2 \times 4^{3/2} \int_{-\pi/2}^{\pi/2} (1 - \sin^2 \theta)^{3/2} \cos \theta d\theta$$

$$= 2 \times 8 \int_{-\pi/2}^{\pi/2} (\cos^2 \theta)^{3/2} \cos \theta d\theta$$

$$= 16 \int_{-\pi/2}^{\pi/2} \cos^3 \theta \cos \theta d\theta = 16 \int_{-\pi/2}^{\pi/2} \cos^4 \theta d\theta$$

$$= 16 \int_{-\pi/2}^{\pi/2} (\cos^2 \theta)^2 d\theta = 16 \int_{-\pi/2}^{\pi/2} \left(\frac{1 + \cos 2\theta}{2} \right)^2 d\theta$$

$$= \frac{16}{4} \int_{-\pi/2}^{\pi/2} (1 + 2 \cos 2\theta + \cos^2 2\theta) d\theta$$

$$= 4 \int_{-\pi/2}^{\pi/2} \left(1 + 2 \cos 2\theta + \frac{1 + \cos 4\theta}{2} \right) d\theta$$

$$= 2 \int_{-\pi/2}^{\pi/2} (2 + 4 \cos 2\theta + 1 + \cos 4\theta) d\theta$$

$$= 2 \int_{-\pi/2}^{\pi/2} (3 + 4 \cos 2\theta + \cos 4\theta) d\theta$$

$$= 2 \left[3\theta + 4 \frac{\sin 2\theta}{2} + \frac{\sin 4\theta}{4} \right]_{-\pi/2}^{\pi/2}$$

$$= 2 \left[3\left(\frac{\pi}{2} + \frac{\pi}{2}\right) + 2(\sin \pi + \sin \pi) + \frac{1}{4}(\sin 2\pi + \sin 2\pi) \right]$$

$$= 2 \times 3 \times \pi + 0 + 0 = 6\pi$$

Ex $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$

$$= \int_0^1 \int_0^1 \left(x^2 z + y^2 z + \frac{z^3}{3} \right) \Big|_0^1 dy dx$$

$$= \int_0^1 \int_0^1 (x^2 + y^2 + 1/3) dy dx$$

$$= \int_0^1 \left(x^2 y + \frac{y^3}{3} + \frac{1}{3} y \right) \Big|_0^1 dx$$

$$= \int_0^1 \left(x^2 + \frac{1}{3} + \frac{1}{3} \right) dx = \left[\frac{x^3}{3} + \frac{1}{3} x + \frac{1}{3} x \right]_0^1$$

$$= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 3/3 = 1$$

Ex $\int_1^e \int_1^e \int_1^e \frac{1}{xyz} dx dy dz$ S3

$$\begin{aligned} \int_1^e \int_1^e \frac{1}{yz} \ln x \Big|_1^e &= \int_1^e \int_1^e \frac{1}{yz} dy dz & \ln e = 1 \\ &= \int_1^e \frac{1}{z} \ln y \Big|_1^e dz & \ln 1 = 0 \\ &= \int_1^e \frac{1}{z} dz & \ln e = 1 \\ &= \ln z \Big|_1^e & \ln 1 = 0 \\ &= \ln e - \ln 1 \\ &= 1 - 0 = 1 \end{aligned}$$

Average Value of a function in space.

The average value of a function f over a region D in space is defined by the formula

$$\text{Average Value} = \frac{1}{\text{Volume of } D} \iiint_D f \, dV$$

Ex Find the average value of $F(x, y, z) = xyz$ over the cube bounded by the coordinate planes and the planes $x=2$, $y=2$ and $z=2$ in the first octant.

Sol The volume of cube $= xyz = 2 \times 2 \times 2 = 8$

$$\int_0^2 \int_0^2 \int_0^2 xyz \, dz \, dy \, dx$$

$$= \int_0^2 \int_0^2 xy \left[\frac{z^2}{2} \right]_0^2 \, dy \, dx = \int_0^2 \int_0^2 xy \left(\frac{2^2 - 0}{2} \right) \, dy \, dx$$

$$= 2 \int_0^2 \int_0^2 xy \, dy \, dx$$

$$= 2 \int_0^2 x \left[\frac{y^2}{2} \right]_0^2 \, dx$$

$$= 2 \times 2 \int_0^2 x \, dx = 4 \left[\frac{x^2}{2} \right]_0^2$$

$$= 4 \times 2 = 8$$

Average value of xyz over the cube $= \frac{1}{\text{Volume}} \iiint_{\text{Cube}} xyz \, dV$

$$= \frac{1}{8} \times 8 = 1$$

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Exercises 1 — 10

Triple integrals in cylindrical and Spherical coordinates

Equations relating rectangular (x, y, z) and cylindrical (r, θ, z) coordinates

$$x = r \cos \theta \quad r^2 = x^2 + y^2$$

$$y = r \sin \theta \quad \theta = \tan^{-1} y/x$$

$$dV = dz \, r \, dr \, d\theta \quad z = z$$

Equations relating rectangular (x, y, z) and spherical (ρ, ϕ, θ) coordinates

$$x = \rho \cos \theta \sin \phi$$

$$y = \rho \sin \theta \sin \phi$$

$$z = \rho \cos \phi$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$E_x \int_0^{2\pi} \int_0^{\theta/2\pi} \int_0^{3+24z^2} dz \, z \, dz \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\theta/2\pi} z \Big|_0^{3+24z^2} dz \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\theta/2\pi} (3+24z^2) z \, dz \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\theta/2\pi} (3z + 24z^3) dz \, d\theta$$

$$= \int_0^{2\pi} \left(\frac{3z^2}{2} + \frac{24z^4}{4} \right) \Big|_0^{\theta/2\pi} d\theta$$

$$= \int_0^{2\pi} \left(\frac{3}{2} \frac{\theta^2}{4\pi^2} + \frac{6}{16\pi^4} \right) d\theta$$

$$= \frac{3}{2 \times 4\pi^2} \frac{\theta^3}{3} + \frac{6}{16\pi^4} \frac{\theta^5}{5} \Big|_0^{2\pi}$$

$$= \frac{1}{8\pi^2} \times 8\pi^3 + \frac{6}{16\pi^4} \frac{32\pi^5}{5}$$

$$= \pi + \frac{12}{5}\pi = \frac{17}{5}\pi$$

Evaluate $\int_0^\pi \int_0^\pi \int_0^{2\sin\phi} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$

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$$= \int_0^\pi \int_0^\pi \left. \frac{\rho^3}{3} \sin\phi \right|_0^{2\sin\phi} d\phi \, d\theta$$

$$= \int_0^\pi \int_0^\pi \frac{8 \sin^3\phi \sin\phi}{3} d\phi \, d\theta$$

$$= \frac{8}{3} \int_0^\pi \int_0^\pi \sin^4\phi \, d\phi \, d\theta$$

$$\sin^2\phi = \frac{1 - \cos 2\phi}{2}$$

$$(\sin^2\phi)^2 = \frac{(1 - \cos 2\phi)^2}{2}$$

$$= \frac{1 - 2\cos 2\phi + \cos^2 2\phi}{2}$$

$$= \frac{1 - 2\cos 2\phi + \frac{1 + \cos 4\phi}{2}}{2}$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{\cos 4\phi}{4} - \frac{2\cos 2\phi}{2}$$

$$\int_0^\pi \sin^4\phi \, d\phi$$

$$= \left. \frac{3}{4}\phi + \frac{\sin 4\phi}{4 \times 4} - \frac{2\sin 2\phi}{2 \times 2} \right|_0^\pi$$

$$= \frac{3}{4}\pi + 0 - 0 = \frac{3}{4}\pi$$

$$= \frac{8}{3} \times \frac{3}{4} \pi \int_0^\pi d\theta = \underline{\underline{2\pi}}$$